

# Early-Stage Cyber Risk Assessment Using Bayesian Networks on CICIDS-2017

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**Abstract**—This paper presents a Bayesian-based probabilistic risk assessment model designed to identify potential cyber breaches within a 24-hour window using the CICIDS-2017 dataset. We focus on low-severity features such as login time anomalies, abnormal port access, and traffic characteristics. The Bayesian network is trained using weak signal indicators and evaluated through variable elimination inference. Results show that even subtle features can collectively yield a high-probability breach prediction, highlighting the utility of Bayesian reasoning in proactive threat detection.

**Index Terms**—Bayesian Network, CICIDS-2017, Cybersecurity, Risk Assessment, Weak Signals, Intrusion Detection, Probabilistic Inference

## I. INTRODUCTION

Traditional intrusion detection systems (IDS) often rely on known attack signatures or high-severity anomalies. However, advanced persistent threats (APT) tend to start with weak or non-obvious signals. This research explores the application of Bayesian networks to perform early-stage risk assessment using weak indicators from CICIDS-2017, one of the most comprehensive public datasets simulating real-world attacks [1].

## II. RESEARCH OBJECTIVE

The primary objective is to determine whether Bayesian networks can effectively estimate the likelihood of a breach within 24 hours based on subtle indicators. The approach involves:

- Extracting weak signals from flow-based traffic logs.
- Designing a Bayesian Network structure.
- Estimating risk probabilities using probabilistic inference.

## III. DATASET DESCRIPTION

The Canadian Institute for Cybersecurity's CICIDS-2017 dataset [1] was used, focusing on the "Friday-WorkingHours-Afternoon-DDoS.pcap" CSV file. This dataset includes simulated DDoS traffic during working hours with over 79 features per flow.

## IV. FEATURE ENGINEERING

Four binary indicators were created:

- **Is\_Off\_Hours**: Access during non-standard hours.
- **Short\_Duration**: Flow duration below 50,000 ms.
- **Low\_Byte\_High\_Packet**: Small byte transfers with excessive packets.

- **Suspicious\_Port**: Use of vulnerable ports (21, 22, 8080).

A synthetic binary label *BreachWithin24h* was generated from the combination of these features to simulate risk.

```
1: import pandas as pd
2: df = pd.read_csv("cicids_bayesian_clean.csv")
3: df.head()
4:
5:   Is_Off_Hours  Short_Duration  Low_Byte_High_Packet  Suspicious_Port  BreachWithin24h
6: 0            0              1                    0              0              0
7: 1            0              1                    0              0              0
8: 2            0              1                    0              0              0
9: 3            0              1                    0              0              0
10: 4            0              1                    0              0              0
11:
12: df.columns = ["Is_Off_Hours", "Short_Duration", "Low_Byte_High_Packet", "Suspicious_Port", "BreachWithin24h"]
```

## V. BAYESIAN NETWORK DESIGN

A Bayesian Network was constructed using the pgmpy library [2], where the selected features were modeled as parent nodes to the breach outcome.

### A. Structure

- $\text{Is\_Off\_Hours} \rightarrow \text{BreachWithin24h}$
- $\text{Short\_Duration} \rightarrow \text{BreachWithin24h}$
- $\text{Low\_Byte\_High\_Packet} \rightarrow \text{BreachWithin24h}$
- $\text{Suspicious\_Port} \rightarrow \text{BreachWithin24h}$

The model was trained using the Bayesian Estimator with a BDeu prior.

## VI. INFERENCE AND RESULTS

Variable Elimination was used for probabilistic inference. An example input scenario is:

```
Is_Off_Hours = 0
Short_Duration = 1
Low_Byte_High_Packet = 0
Suspicious_Port = 1
```

The output probability was:

- $P(\text{BreachWithin24h} = 1) = 0.9976$
- $P(\text{BreachWithin24h} = 0) = 0.0024$

This demonstrates the model's sensitivity to subtle threat patterns.

```

from pgmpy.inference import VariableElimination

inference = VariableElimination(model)

result = inference.query(
    variables=['BreachWithin24h'],
    evidence={
        'Is_Off_Hours': df['Is_Off_Hours'].cat.categories[0],
        'Short_Duration': df['Short_Duration'].cat.categories[1],
        'Low_Byte_High_Packet': df['Low_Byte_High_Packet'].cat.categories[0],
        'Suspicious_Port': df['Suspicious_Port'].cat.categories[1]
    }
)

print(result)

```

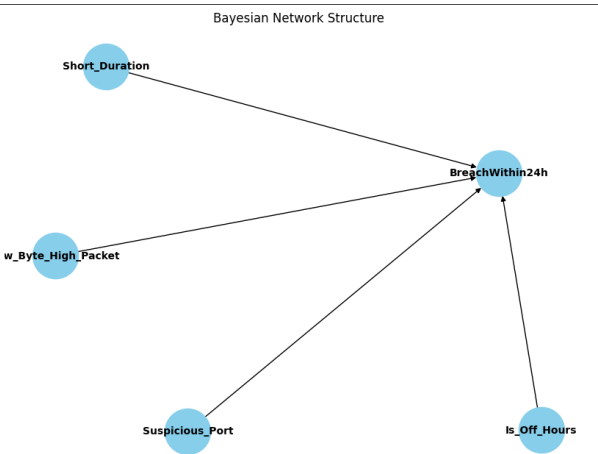
| BreachWithin24h    | phi(BreachWithin24h) |
|--------------------|----------------------|
| BreachWithin24h(0) | 0.0024               |
| BreachWithin24h(1) | 0.9976               |

and intrusion traffic characterization,” *ICISSP*, 2018.

[2] A. Das and S. Mukherjee, ”pgmpy: Probabilistic Graphical Models using Python,” arXiv preprint arXiv:1409.3273, 2015.

### VII. GRAPHICAL REPRESENTATION

The model was visualized using `networkx`. Edges in the graph represent the direct influence of each feature on breach probability. This enhances explainability, allowing analysts to trace the reasoning process.



### VIII. CONCLUSION

This research confirms that Bayesian networks can successfully assess cybersecurity risk based on weak signals in flow-based network logs. Unlike deterministic models, Bayesian inference offers probabilistic, explainable predictions suitable for integration into modern SIEM systems.

#### Future Work

Future directions include:

- Temporal modeling using Dynamic Bayesian Networks (DBNs).
- Real-time stream integration for automated alerting.
- Evaluation with diverse attack types across the CICIDS dataset.

#### References

[1] M. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, ”Toward generating a new intrusion detection dataset

